

But it still seems one of history's more mischievous twists that a goal that had eluded distinguished scientists and engineers, as well as rich enthusiasts, should have been attained by two brothers who ran a bicycle shop in Dayton, Ohio.

Men of their time

Although they lived far from the centers of power and fashion, Orville and Wilbur Wright had grown up very much in touch with contemporary currents of thought and innovation. Their formative years were a time when new inventions proliferated – the telephone, automobiles, electric light, wireless telegraphy, and the movies. Inventors such as Thomas Edison and Alexander Graham Bell were the heroes of the age. It would have been surprising had the Wrights not taken some interest in the widely publicized flight experiments of the 1890s. Interest seems to have blossomed into committed research through the perception that flight was, as Wilbur wrote, “almost the only great problem that has not been... carried to a point where further progress is very difficult.”

From the outset, the Wrights displayed the systematic approach that was to characterize their entire endeavor. Their first need was to absorb

existing knowledge. In May 1899, Wilbur wrote a letter to the Smithsonian Institution, asking for any papers it might have on flight and a reading list of books on the subject. “I wish to avail myself of all that is already known,” Wilbur wrote, “and then if possible add my mite to help on the future worker who will attain final success.” The letter received a prompt and helpful response.

When the Wrights had acquainted themselves with the works of, among others, Cayley, Pénaud, Chanute, Lilienthal, and Langley, they identified an area that seemed to have been neglected: control. Men such as Langley had imagined a flying machine to be rather like a car – an essentially stable machine to be switched on and then steered. However, the Wrights instinctively

felt that a flying machine was more like a bicycle, and would need to be flown with constant adjustments of balance. From the start they posed the problem not simply of how to build a flying machine, but also how to fly it.

“... I have some pet theories as to the proper construction of a flying machine...”

WILBUR WRIGHT

LETTER TO THE SMITHSONIAN INSTITUTION, 1899



LETTER TO THE SMITHSONIAN

In 1899 Wilbur Wright wrote to the Smithsonian, expressing his interest in aviation and asking for a reading list: “I wish to avail myself of all that is already known and then if possible add my mite to help on the future worker who will attain final success.”

First experiments

Their first breakthrough came from a more traditional direction. Watching soaring buzzards, Wilbur was struck by the movement of the feathers on their wingtips, which kept the birds' lateral balance. The brothers puzzled for a long time over achieving a similar effect on an aircraft wing, until Wilbur had a sudden moment of inspiration. Absent-mindedly twisting the ends of a narrow cardboard box in opposite directions, he saw that the same could be done with a wing. “Wing warping” had been devised.

By 1900, the Wrights were ready to begin experiments with a glider, which they built in their bicycle workshop. Glider experiments required wind, and having contacted the US Weather Bureau, they established that Kitty Hawk, a small beach settlement on the coast of North Carolina, would provide

TESTING THE GLIDER

Intended as a man-carrying glider, this kite-glider (their second) had a wingspan of 17ft (5.2m) and a forward elevator. However, tests at Kill Devil Hills, near Kitty Hawk, in September 1900, soon revealed that the efficiency of the airfoil was insufficient to support a man's weight, unless the wind was very strong.

